Rave Generation

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SYMPHONIC

MULTI-FX PROCESSOR

User Manual

1. Overview

Symphonic runs the Yamaha SPX90 Multi-Effects Processor's own ROM: its original firmware and DSP programs, at the hardware's native 31.25 kHz internal clock with a 12 kHz passband. All 30 factory presets are here: four reverbs, early reflections, delay and echo, the seven modulation effects, gated reverbs, a noise gate and a compressor, pitch change, freeze, auto pan, delay vibrato and a parametric EQ.

The panel works like the unit's. The LCD shows the preset and the parameter you are setting, in the SPX90's own words; the red MEMORY display shows the preset number; the LEDs show the input level. Projects made with Symphonic 1.0 open with their settings (section 8).

2. Signal Flow

Input Gain → Anti-Alias Filter (12 kHz) → SRC to 31.25 kHz → SPX90 Effect and Balance → SRC to Host Rate → DAC Reconstruction Filter (12 kHz) → Output Gain

Audio enters the plugin and is scaled by the In Lvl control, then passed through a 12 kHz anti-alias low-pass that mirrors the original hardware's analog input stage. The signal is converted to the SPX90's native 31.25 kHz sample rate and 16-bit resolution, then fed to the selected effect.

The SPX90 hardware is mono-in, stereo-out: stereo input is summed to mono before entering the effect, and the stereo image is created by the effect itself. After processing, the output is resampled back to the host session rate and passed through a 12 kHz reconstruction filter.

Balance mixes the direct and effect signals inside the SPX90, as on the hardware, so the direct signal passes through its converters too. Out Lvl then sets the final level. The conversion takes 3.1 ms, which Symphonic reports to your DAW so its delay compensation keeps every track in time.

3. The Panel

Everything on the panel works as on the SPX90's front panel, with knobs in place of its PARAMETER keys.

Display

The LCD shows what the SPX90 itself shows: the preset name on the top line and, on the bottom line, the parameter you are setting with its value. FREEZE also shows its status there (REC. READY while it waits, OK when a recording is done). The red MEMORY display shows the preset number, 1 to 30. The LEDs show the level going into the unit's 16-bit converter: keep the red ones dark.

MEMORY and RECALL

The MEMORY ▲ and ▼ keys step through the 30 presets in the unit's order and load each one with its factory values. RECALL puts the factory values of the current preset back on all knobs, which undoes your edits.

Knobs

BALANCE and OUT LVL are on every preset. BALANCE mixes the direct sound and the effect: 0% is direct only, 100% effect only, for use on a send. OUT LVL sets the unit's output level. The other knobs are the preset's own parameters, named as on the LCD and in the order the SPX90's PARAMETER key steps through them; knobs a preset does not use are hidden. Each knob shows its value underneath, and turning one shows it on the LCD.

Keys

BYPASS passes the input through unprocessed; its LED lights while it is on. TRIGGER is the SPX90's footswitch input: it fires the noise gate, compressor, reverb and gate, triggered pan and delay vibrato, and starts a FREEZE recording. RECORD, OVER DUB and PLAY operate FREEZE A and B. Keys the current preset does not use are greyed out.

INPUT and OUTPUT

INPUT sets the level into the unit, OUTPUT the level out of the plugin, each from -20 to +6 dB. Set INPUT by the LEDs: the SPX90's converter clips above its top LED.

MIDI

Symphonic listens to MIDI notes on every channel, as the SPX90 does on its MIDI IN. With MIDI TRG. ON, a note fires the noise gate, compressor, reverb and gate, triggered pan or delay vibrato. On PITCH CHANGE A and D a note sets the pitch, counted from BASE KEY; on FREEZE B it plays the recording at that pitch. Send a MIDI track to Symphonic the way your DAW allows for effect plugins.

Hints

Hover over a control to read what it does in the bar at the bottom of the window. The (i) icon at the top right turns the hints on and off.

4. The 30 Presets

The presets are the SPX90's factory memories 1 to 30, in the unit's order. In your DAW's preset list the seven modulation presets (9 to 15) come first, so projects made with Symphonic 1.0 keep their settings.

1 to 4 Reverb

REV 1 HALL, REV 2 ROOM, REV 3 VOCAL and REV 4 PLATE: a large hall, a room, a reverb voiced for vocals and a bright plate. All four have the same parameters; their factory settings give each its character.

Parameter	Range	Description
REV TIME	0.3 to 99.0 s	Reverb time: how long the reverb takes to fade by 60 dB at 1 kHz.
HIGH	0.1 to 1.0	Decay of the high frequencies relative to REV TIME. Lower values give a darker tail.
DELAY	0.1 to 50.0 ms	Time between the direct sound and the start of the reverb.
HPF	THRU, 32 Hz to 1.0 kHz	Low cut on the reverb. THRU is off.
LPF	1.0 kHz to 11 kHz, THRU	High cut on the reverb. THRU is off.

5 and 6 Early Reflections

The first reflections of a space without the reverb tail, for presence and size without wash. EARLY REF. 2 has more reflections than EARLY REF. 1.

Parameter	Range	Description
TYPE	HALL, RANDOM, REVERSE, PLATE	Pattern of the reflections: as in a hall or a plate, an irregular RANDOM series, or REVERSE, rising in level like a reverb played backwards.
ROOM SIZE	0.1 to 20.0	Spacing of the reflections: the size of the room.
LIVENESS	0 to 10	How slowly the reflections fade: 0 is a dead room, 10 a hard, reflective one.
DELAY	0.1 to 400.0 ms	Time between the direct sound and the first reflection.
LPF	1.0 kHz to 11 kHz, THRU	High cut. THRU is off.

7 and 8 Delay and Echo

Separate delays for the left and right outputs, each with its own feedback. DELAY L,R reaches 500 ms per side; STEREO ECHO reaches 250 ms and is set up for long runs of repeats.

Parameter	Range	Description
Lch DLY, Rch DLY	0.1 to 500.0 ms (ECHO: 250.0 ms)	Left and right delay time.
Lch F.B, Rch F.B	-99 to +99%	Feedback: more repeats as it rises. Negative values invert the repeats' polarity.
HIGH	0.1 to 1.0	High frequencies in the feedback. Lower values give darker repeats.

9 Stereo Flange A

A classic flanger using a short, LFO-modulated delay. MOD. DLY sets the base delay time, MOD. FRQ the sweep rate and MOD. DEPTH the sweep width. F.B. GAIN feeds the delayed signal back into the input for a sharper, more resonant comb-filter character. The two output channels sweep 202.5 degrees apart, giving a slightly asymmetric, lively stereo movement. Factory settings: MOD. FRQ 2.5 Hz, MOD. DEPTH 50%, MOD. DLY 1.2 ms, F.B. GAIN 35%, BALANCE 50%.

10 Stereo Flange B

The same flanger as Flange A with a slow, deep factory setting: a 0.5 Hz sweep at 90% depth with a little more feedback, for a broad jet sweep. Factory settings: MOD. FRQ 0.5 Hz, MOD. DEPTH 90%, MOD. DLY 1.0 ms, F.B. GAIN 40%, BALANCE 75%.

11 Chorus A

A warm, thick chorus combining delay modulation (MOD. FRQ and DM DEPTH) with simultaneous amplitude modulation of the wet signal (AM DEPTH). This dual-modulation approach gives Chorus A its characteristic organic shimmer. Factory settings: MOD. FRQ 0.2 Hz, DM DEPTH 50%, AM DEPTH 40%.

12 Chorus B

A brighter, more forward chorus. It uses the same delay-and-amplitude modulation as Chorus A but moves less, giving a subtler shimmer. Use Chorus B when you want the thickening quality of chorus without as much of the amplitude pulsing. Factory settings: MOD. FRQ 0.6 Hz, DM DEPTH 50%, AM DEPTH 10%.

13 Stereo Phasing

A phaser built on a short modulated delay, swept by the SPX90's LFO for its characteristic swooping, hollow sound. Its MOD. DLY covers the unit's phasing range of 0.1 to 8 ms. Factory settings: MOD. FRQ 1.1 Hz, MOD. DEPTH 100%, MOD. DLY 3.0 ms.

14 Tremolo

A pulsing amplitude modulator driven by the SPX90's original LFO. MOD. FRQ sets the rate and MOD. DEPTH how far the level dips in each cycle: at the factory 50% it almost cuts out between pulses. As on the hardware, Tremolo plays about 6 dB louder than the other presets; lower OUT LVL or OUTPUT to match. Factory settings: MOD. FRQ 6.0 Hz, MOD. DEPTH 50%.

15 Symphonic

The Symphonic patch is the most distinctive effect in the SPX90's modulation bank. Where the chorus and flanger move a single delayed voice, Symphonic layers several, each drifting at a different point in the modulation cycle. One voice stays common to both outputs while the others separate left and right, and that is what opens the sound out into a wide ensemble rather than an obvious chorus. The movement is slow and shallow by design: it thickens and widens without the pitch wobble of a conventional chorus. Factory settings: MOD. FRQ 0.7 Hz, MOD. DEPTH 50%.

The modulation parameters:

Parameter	Range	Description
MOD. FRQ	0.1 to 20.0 Hz	Speed of the modulation.
MOD. DEPTH, DM DEPTH	0 to 100%	Amount of modulation: the sweep width, or for TREMOLO the level change. At 100% the delay swings by up to 4 ms.
MOD. DLY	0.1 to 100.0 ms (PHASING: 0.1 to 8.0 ms)	Base delay. Below 1 ms the sweep moves in the high frequencies, above 3 ms in the low ones.
F.B. GAIN	0 to 99%	Feedback, for a stronger and more resonant flange (FLANGE A and B).
AM DEPTH	0 to 100%	Level modulation on top of the delay modulation (CHORUS A and B).

16 and 17 Gate Reverb and Reverse Gate

Early reflections cut off sharply: the gated sound of 1980s drums. REVERSE GATE uses the REVERSE pattern, so the sound swells before it stops. The parameters are those of the Early Reflections (TYPE, ROOM SIZE, LIVENESS, DELAY, LPF).

18 ADR-Noise Gate

A gate opened by the input level or by TRIGGER, with its own attack, decay, hold and release. Use it as a noise gate, to cut sounds to a set length, or with a slow ATTACK for a reverse swell.

Parameter	Range	Description
TRG. LEVEL	1 to 100	Input level that fires the effect. Higher values need a louder input; TRIGGER always fires it.
TRG. DLY	-100 to +100 ms	Time from the trigger to the start of the effect. Negative values delay the input instead, so the effect starts before the sound.
TRG. MSK	5 to 32000 ms	After a trigger, new triggers are ignored for this long.
ATTACK	5 to 32000 ms	Time for the gate to open fully.
DECAY	5 to 32000 ms	Time to fall to DECAY LVL once open.
DECAY LVL	0 to 100%	Level the gate holds during HOLD.
HOLD	1 to 30000 ms	How long the gate stays open.
RELEASE	5 to 32000 ms	Time for the gate to close.
MIDI TRG.	OFF, ON	ON: a MIDI note fires it too.

19 Compressor

Pulls the level of each attack down to HOLD LEVEL for the HOLD time, then lets it back up over RELEASE. It is triggered like the gate.

Parameter	Range	Description
TRG. LEVEL	1 to 100	Input level that fires the effect. Higher values need a louder input; TRIGGER always fires it.
TRG. DLY	-100 to +100 ms	Time from the trigger to the start of the effect. Negative values delay the input instead, so the effect starts before the sound.

Parameter	Range	Description
TRG. MSK	5 to 32000 ms	After a trigger, new triggers are ignored for this long.
ATTACK	5 to 32000 ms	Time to reach HOLD LEVEL after a trigger.
HOLD	1 to 30000 ms	How long the compression is held.
HOLD LEVEL	0 to 100%	Level during HOLD. Lower values compress harder.
RELEASE	5 to 32000 ms	Time to return to the normal level.
MIDI TRG.	OFF, ON	ON: a MIDI note fires it too.

20 Reverb & Gate

A reverb whose output passes through a gate: the input opens it, it stays open for HOLD and closes over RELEASE. REV TIME, HIGH, DELAY, HPF and LPF work as in presets 1 to 4.

Parameter	Range	Description
TRG. LEVEL	1 to 100	Input level that opens the gate. Higher values need a louder input; TRIGGER always opens it.
HOLD	1 to 30000 ms	How long the gate stays open.
RELEASE	5 to 32000 ms	Time for the gate to close.
MIDI TRG.	OFF, ON	ON: a MIDI note fires it too.

21 to 24 Pitch Change

Shifts the input up or down by as much as an octave. A and D give one shifted voice with feedback, for repeats that climb or fall further each time. B gives two voices in the centre (1 and 2), C one on each side (L and R): with a few cents of FINE they make a thick doubling, with larger steps a harmony.

Parameter	Range	Description
PITCH	-12 to +12	Pitch shift in semitones.
FINE	-100 to +100	Fine shift in cents.

Parameter	Range	Description
DELAY, DLY	0.1 to 400.0 ms (C: 200.0 ms)	Delay of the shifted voice.
F.B. GAIN	0 to 99%	Repeats of the shifted voice, each shifted again (A and D).
BASE KEY	OFF, C 1 to C 6	A and D: a MIDI note sets PITCH, counted from this key (C 3 is middle C, note 60). OFF ignores notes.

25 and 26 Freeze

Records up to 500 ms of the input and plays it back on demand. FREEZE A can play part of the recording (START, END); FREEZE B plays it back at another pitch (PITCH, FINE).

To record, choose REC MODE and press RECORD. The LCD shows REC. READY. In MANUAL the recording starts at once; in AUTO it starts when the input gets loud enough or when you press TRIGGER. The LCD shows OK when the 500 ms are in. OVER DUB records on top of what is already there, the same way. PLAY plays the recording, once per press.

Parameter	Range	Description
REC MODE	MANUAL, AUTO	MANUAL records as soon as you press RECORD; AUTO waits for the input or TRIGGER.
TRG. DLY	-500 to +500 ms	When the recording starts relative to the trigger. Negative values start it before the trigger, so the attack is not lost.
START, END	0 to 500 (ms)	The part of the recording FREEZE A plays.
PITCH, FINE	-12 to +12, -100 to +100	Playback pitch in semitones and cents (FREEZE B).
INPUT TRG.	OFF, ON	FREEZE A: ON plays the recording each time the input gets loud enough or TRIGGER is pressed.
BASE KEY	OFF, C 1 to C 6	FREEZE B: a MIDI note plays the recording, pitched from this key.

27 and 28 Pan and Triggered Pan

PAN moves the sound between the outputs all the time; TRIGGERED PAN makes one move each time the input or TRIGGER fires it.

Parameter	Range	Description
PAN SPEED	0.1 to 20.0 Hz	Pan rate (PAN).

Parameter	Range	Description
DIRECTION	L→R, L←R, L↔R (TRIGGERED PAN: L→R, L←R)	Direction of the movement.
DEPTH	0 to 100%	How far the sound moves (PAN).
TRG. LEVEL, TRG. DLY, TRG. MSK	as on the noise gate	When TRIGGERED PAN fires.
ATTACK	5 to 32000 ms	Time to reach the full pan position.
PANNING	5 to 32000 ms	Time the pan stays at full depth.
RELEASE	5 to 32000 ms	Time to come back.
L/R BALANCE	0 to 100%	Level balance between the left and right outputs.
MIDI TRG.	OFF, ON	ON: a MIDI note fires it too.

29 Delay Vibrato

A vibrato that waits and then fades in each time the input or TRIGGER fires it, like a singer's or a string player's delayed vibrato.

Parameter	Range	Description
TRG. LEVEL	1 to 100	Input level that restarts the vibrato. Higher values need a louder input.
VIB DLY	1 to 30000 ms	Time without vibrato after a trigger.
VIB RISE	5 to 32000 ms	Time for the vibrato to reach full depth.
VIB FRQ	0.1 to 20.0 Hz	Vibrato rate.
VIB DEPTH	0 to 100%	Vibrato depth.
MIDI TRG.	OFF, ON	ON: a MIDI note fires it too.

30 Parametric EQ

Two parametric bands between a low cut and a high cut, with a delay.

Parameter	Range	Description
HPF	THRU, 32 Hz to 1.0 kHz	Low cut, 6 dB per octave. THRU is off.
MID FRQ	315 Hz to 4.0 kHz	Mid band frequency, in sixth-octave steps.
MID GAIN	-15 to +15 dB	Mid band boost or cut.
MID Q	0.5 to 5.0	Mid band width. Higher values are narrower.
HI FRQ	800 Hz to 8.0 kHz	High band frequency.
HI GAIN	-15 to +15 dB	High band boost or cut.
HI Q	0.5 to 5.0	High band width. Higher values are narrower.
LPF	1.0 kHz to 11 kHz, THRU	High cut. THRU is off.
DELAY	0.1 to 400.0 ms	Delay of the equalised signal.

5. Controls Reference

5.1 Preset

The Preset parameter selects one of the 30 presets. MEMORY ▲ and ▼ and RECALL load factory values; changing the preset from your DAW keeps the knobs where they are.

Control	Range	Description
Preset	30 positions	Selects one of the 30 presets, 1 REV 1 HALL to 30 PARAMETRIC EQ. In the DAW's list the seven modulation presets come first.

5.2 Global Controls

These controls are on every preset.

Control	Range	Description
INPUT (In Lvl)	-20 to +6 dB	Level into the unit. Higher settings drive its 16-bit converter harder; set it by the LEDs.
OUTPUT (Out Lvl)	-20 to +6 dB	Level out of the plugin, after the unit.
BALANCE	0 to 100%	The unit's mix of direct sound and effect: 0% direct only, 100% effect only, for send and return use. As on the hardware, the direct sound also passes through the unit.
OUT LVL	0 to 100%	The unit's output level.
BYPASS	Off / On	Passes the input through unprocessed. The red LED on the key lights while it is on.

5.3 Automation in Your DAW

The seven modulation presets use the parameter names of Symphonic 1.0: Balance, Mod Frq, Mod Depth, Mod Dly, FB Gain, AM Depth, Sym Mod Frq and Sym Mod Depth, plus Phs Mod Dly for STEREO PHASING. The other presets use Unit Balance and Param 1 to Param 9, the knobs from left to right after OUT LVL. Unit Out Lvl is OUT LVL on every preset.

6. Quick Start

Recommended Starting Points

MEMORY ▲ and ▼ load each preset with its factory values, so every one is a usable starting point. The following tips build on those values.

Each sound tip below is also saved as a preset: open the preset menu at the top of the plugin and choose the Starting Points folder. It holds eight more: Slapback Echo, MIDI Note Gate, Drum Room, Vocal Plate, Octave Down, Wide Auto Pan, Telephone Voice and Ambient Hall.

Lush ensemble on strings or pads

Select 15 SYMPHONIC. Leave MOD. FRQ at the factory 0.7 Hz and MOD. DEPTH at 50%. Set BALANCE to 70-80% for a blend that still keeps some of the direct sound. Increase MOD. DEPTH gradually for more shimmer, or lower MOD. FRQ to 0.3-0.5 Hz for a slower, more stately movement.

Classic 1980s flanger on electric guitar or synth

Select 9 STEREO FLANGE A. Its BALANCE of 50% puts the direct and flanged sound side by side for a natural comb effect. Push F.B. GAIN to 50-65% for a more metallic character, raise MOD. FRQ to 3-5 Hz for a faster jet sound, or drop MOD. DLY to 0.5-1.0 ms for a higher, brighter sweep.

Warm chorus on vocals or acoustic guitar

Select 11 CHORUS A, leave BALANCE at 100% and put the plugin on an aux send. The factory MOD. FRQ of 0.2 Hz and AM DEPTH of 40% give a gentle, organic shimmer. Raise AM DEPTH to 60-80% for a more pronounced doubling, or MOD. FRQ to 0.6-1.0 Hz for more movement without losing the width.

Subtle thickening on a mix bus

Select 12 CHORUS B or 15 SYMPHONIC. Set BALANCE to 20-40% for a light blend and lower the depth to 20-30% so the movement is felt rather than heard. It adds dimension and stereo spread without changing the character of the source.

Slow phasing on synth pads

Select 13 STEREO PHASING. Lower MOD. FRQ to 0.3-0.5 Hz and set BALANCE to 80-100%. The phaser sweeps slowly through the harmonics for an evolving, animated sound. Raise MOD. DLY to 4-5 ms if the sweep feels too shallow.

Hall reverb on vocals

Select 1 REV 1 HALL or 3 REV 3 VOCAL on an aux send with BALANCE at 100%. Set REV TIME to 2-3 s and DELAY to 30-60 ms so the voice stays clear of the reverb, and use HPF around 80 Hz to keep the low end out of it.

Gated reverb on a snare

Select 20 REVERB & GATE. Raise TRG. LEVEL until only the snare opens the gate, then set HOLD to 150-300 ms for the length of the burst. 16 GATE REVERB gives a tighter, roomier version.

Vocal doubling

Select 22 PITCH CHANGE B. Its factory FINE values of +8 and -8 cents with a 20 ms second voice make a wide double; set BALANCE to 30-50% on an insert.

Send/return versus insert routing

For the classic send setup, put Symphonic on an aux channel with BALANCE at 100%: your tracks feed it through sends, and the send levels set the amount. On an insert, set BALANCE to taste and use OUT LVL or OUTPUT to match the level.

7. Installation and Troubleshooting

7.1 System Requirements

Before installing Symphonic, ensure your system meets the following requirements:

- Operating system: macOS 10.13 or later, or Windows 10 or later.
- Plugin formats: VST3 and AU.
- Processor: Intel Core i5 or Apple Silicon equivalent.
- RAM: 4 GB minimum (8 GB or more recommended).
- Disk space: 50 MB free.

7.2 Installation

- Download the installer from ravegeneration.io or the platform where you purchased the plugin.
- Run the installer and follow the on-screen instructions.
- Launch your DAW and locate Symphonic in your plugin list.
- If prompted, activate the plugin using the licence key provided with your purchase.

7.3 Troubleshooting

Control	Range	Description
Plugin not showing in DAW		Check that the correct plugin folder path is enabled in your DAW's plugin manager, then rescan the plugin folders. On macOS, VST3 plugins install to /Library/Audio/Plug-Ins/VST3. AU plugins install to /Library/Audio/Plug-Ins/Components.
Activation issues		Verify your internet connection and enter the licence key exactly as provided, including any hyphens. If the problem persists, contact support@ravegeneration.io with your order reference.

8. Projects Made with Symphonic 1.0

Projects saved with Symphonic 1.0 open in this version on the same preset with the same settings. The seven modulation presets now run on the SPX90's own ROM, so they sound like the hardware rather than the 1.0 model; TREMOLO in particular plays about 6 dB louder, as on the unit.

Two 1.0 settings went beyond the unit's limits and now stop at them: Mod Frq above 20 Hz plays at 20 Hz, and a negative FB Gain plays at 0%. STEREO PHASING now has its own MOD. DLY (0.1 to 8 ms), so a 1.0 project on STEREO PHASING comes back with the factory 3.0 ms.

9. Historical Context

The Yamaha SPX90 was released in 1985 as a compact, affordable multi-effects unit aimed at studio engineers and live sound professionals who needed a single rack space to deliver reverb, chorus, flanging, and other effects without the cost of multiple dedicated units. The unit processed audio at 31.25 kHz with a 12 kHz passband, which gave it a noticeably darker, slightly band-limited quality compared to later high-fidelity processors. That quality became part of its sound.

The Symphonic preset in particular became a signature of the era. It appears on recordings across the second half of the 1980s and into the 1990s: on string pads, synth leads, backing vocals, and electric piano. Its layered, independently drifting voices produce a thickening effect that sits between a chorus and a reverb pre-delay, adding spaciousness and width without the obvious pitch wobble of a conventional chorus. Part of what makes Symphonic distinctive is that the modulation is slow enough to be felt rather than heard as separate pitch movement, giving each note an ensemble-like bloom.

The other effects on the SPX90 were equally well-regarded. The flangers were characterised by their feedback resonance and slightly rough edges, the chorus units had a warmth that came partly from the 16-bit processing and the bandwidth limiting, and the tremolo had a smooth, musical pulse. These subtleties are what this plugin is designed to preserve.

Symphonic the plugin is not a sample-based recreation or an impulse-response capture. It runs the SPX90's own firmware and DSP programs at the original 31.25 kHz sample rate, so it responds to parameter changes the way the hardware does.

Support

For technical support, updates, and additional information:

Website: ravegeneration.io

Email: support@ravegeneration.io

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